DELHI TECHNOLOGICAL UNIVERSITY

Multimodal Data Processing (IT-433)



Project:

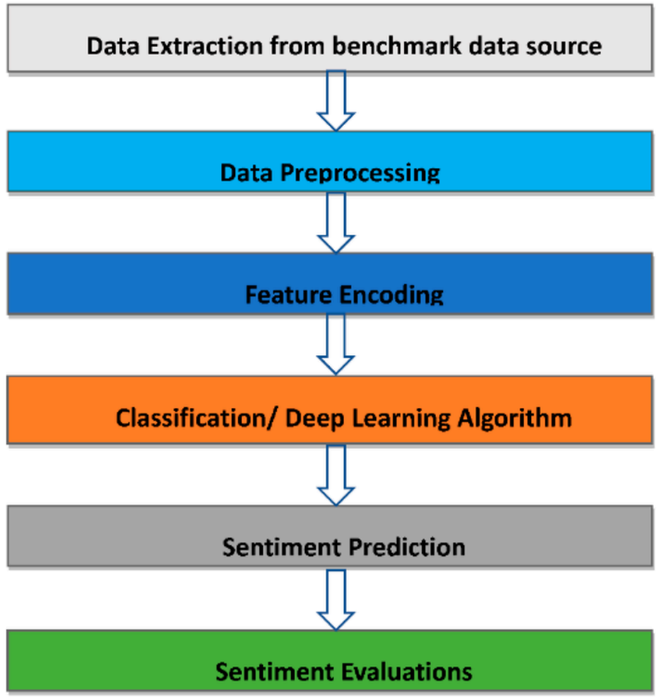
Multimodal Sentiment Analysis Using Text and Images

Submitted to :
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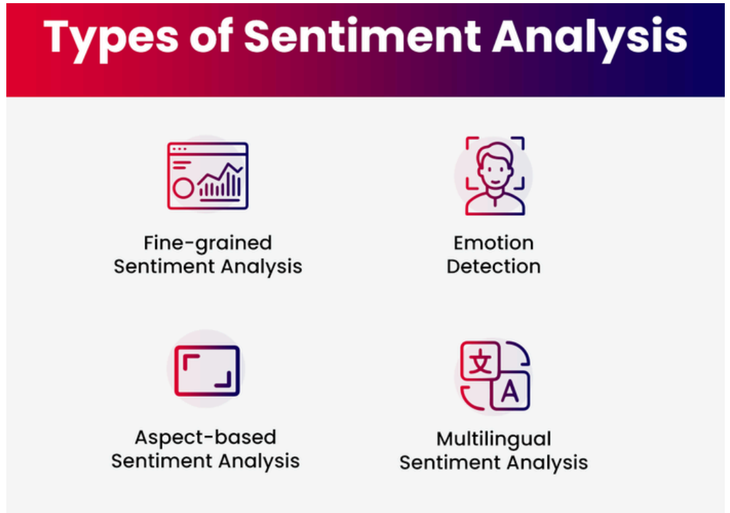
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Abstract

- Multimodal sentiment analysis has become an important research area in the field of artificial intelligence.
- It has great potential for both application and research, making it a popular research topic. This review provides an overview of the definition, background, and development of multi- modal sentiment analysis.
- It also covers recent datasets and advanced models, emphasizing the challenges and future prospects of this technology. Finally, it looks ahead to future research directions.

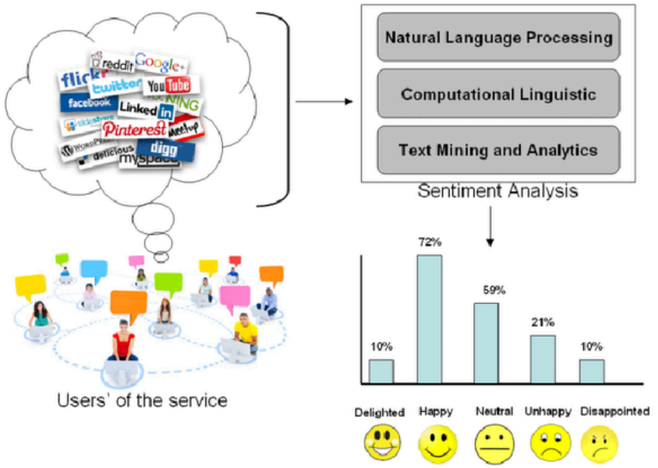


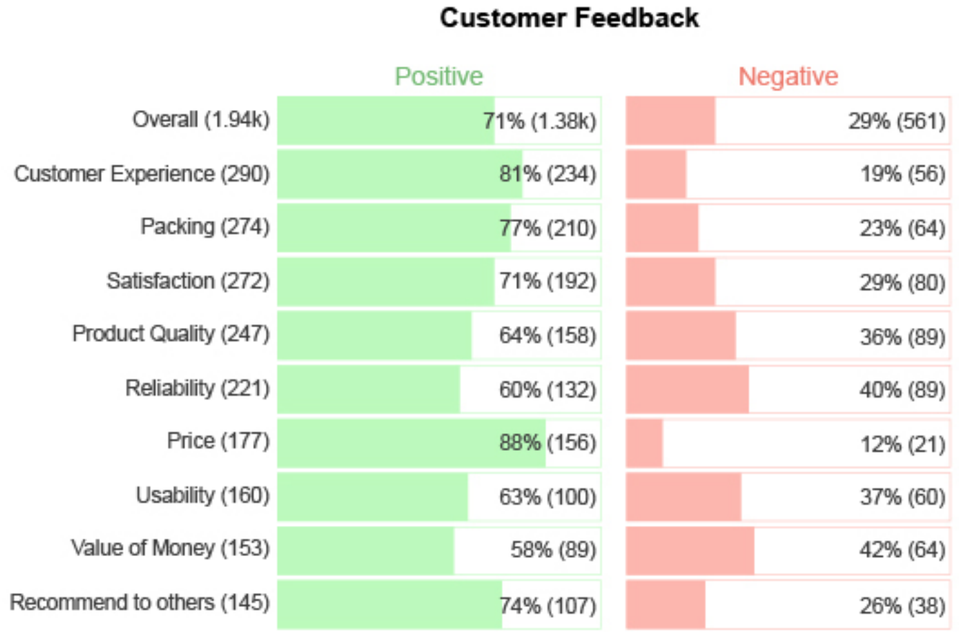
Flowchart of sentiment Analysis



Multimodal Sentiment Analysis

- Sentiment analysis involves analyzing sentiment polarity through available information .
- With the rapid development of fields such as artificial intelligence, computer vision, and natural language processing, it is becoming increasingly possible for artificial agents to implement sentiment analysis.
- Speech-based sentiment analysis mainly extracts features such as pitch, timbre, and temperament in speech for sentiment analysis .







Dataset Introduction

The Enriched MVSA dataset:

[1] D. J. Mohammed and H. J. Aleqabie, "The Enrichment Of MVSA Twitter Data Via Caption-Generated Label Using Sentiment Analysis," 2022 Iraqi International Conference on Communication and Information Technologies (IICCIT), Basrah, Iraq, 2022, pp. 322-327, doi: 10.1109/IICCIT55816.2022.10010435.

The Original Dataset:

[2] T. Niu, S. A. Zhu, L. Pang and A. El Saddik, Sentiment Analysis on Multi-view Social Data, MultiMedia Modeling (MMM), pp: 15-27, Miami, 2016.



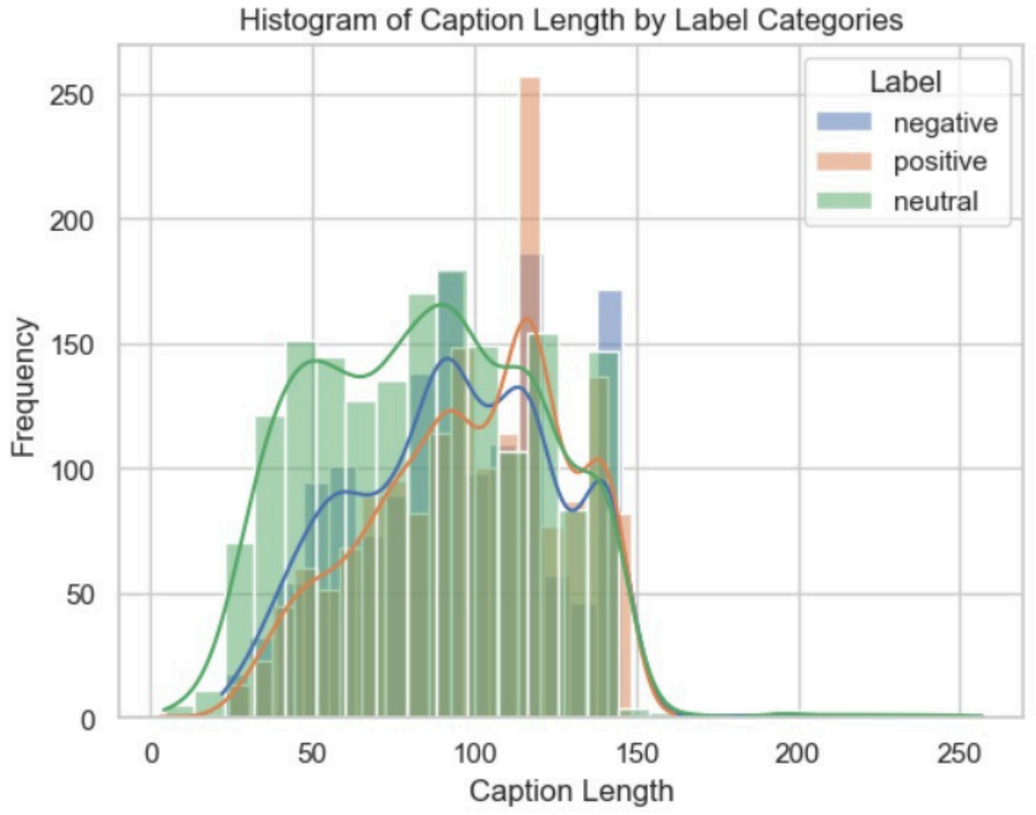
How I feel today #legday #jelly #aching #gym

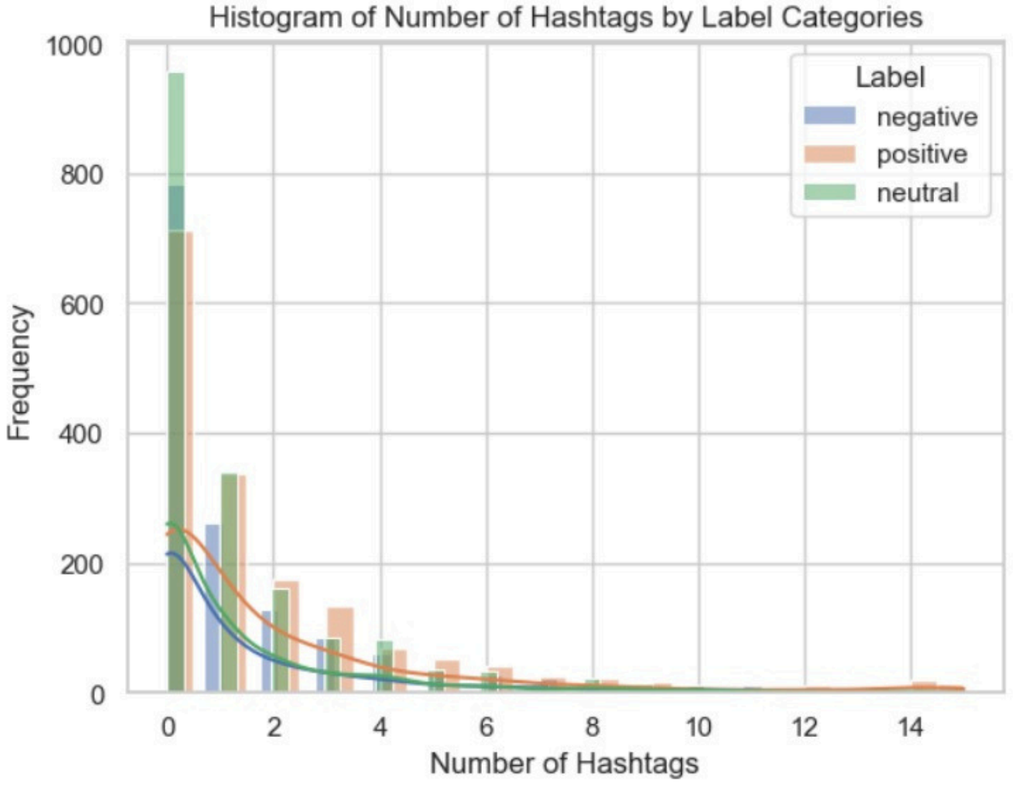
EDA & Feature Engineering

The captions and hashtags are transformed into a more valuable and relevant structure which is easier to process such as:

Caption Length : Indicating length of captions
Hashtags : Extracting and collecting all the hashtags used in each tweet
Total Hashtags : Showing the total number of hashtags in each tweets

Each Tweet is classified into one of 3 sentiments: Positive, Negative and Neutral
Total of ~5000 tweets


Histogram of Caption Length by Label Categories
Label
negative
positive
neutral
Frequency
Caption Length
0
50
100
150
200
250
0
50
100
150
200
250

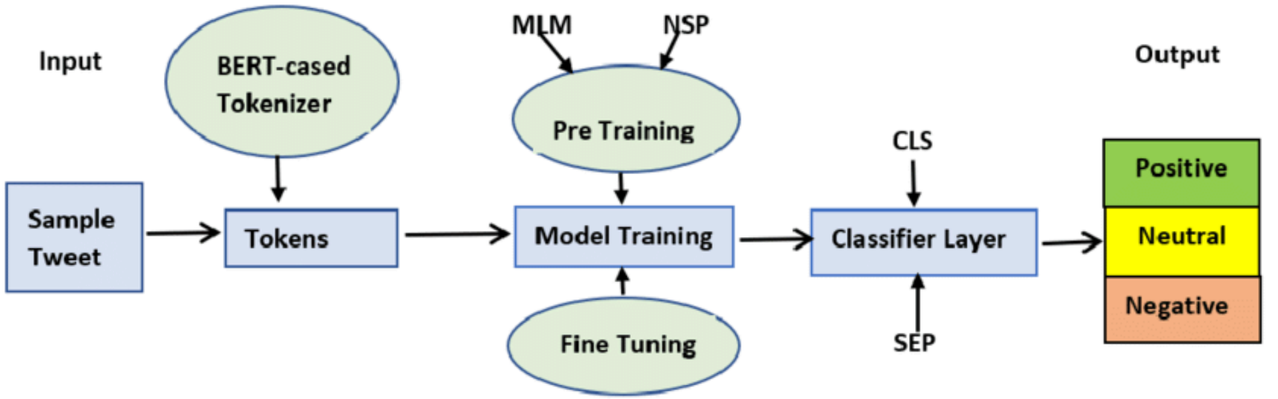

Histogram of Number of Hashtags by Label Categories
Label
negative
positive
neutral
Frequency
Number of Hashtags
0
200
400
600
800
1000
0
2
4
6
8
10
12
14

Generating text embeddings

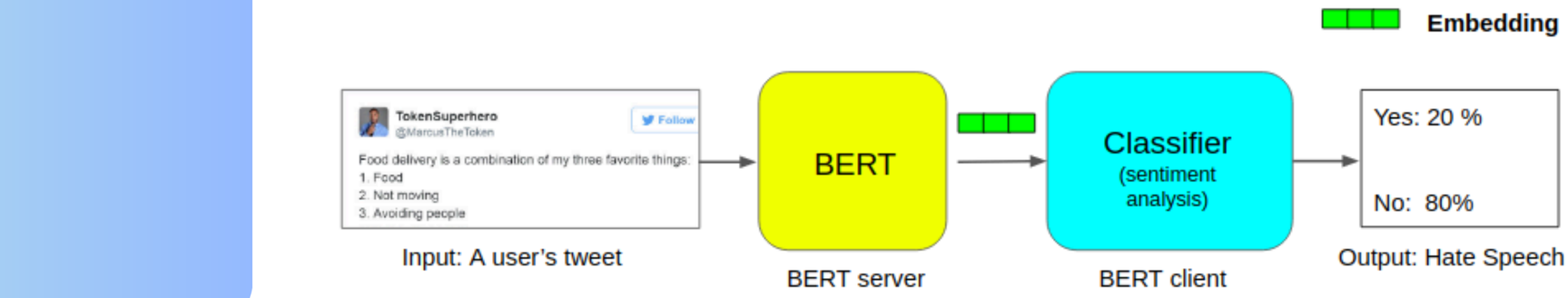
BERT, or Bidirectional Encoder Representations from Transformers, is a natural language processing (NLP) pre-training technique that has revolutionized how we understand and process language.

Key features:-

- **Contextual Understanding**
- **Transformer Architecture**
- **Pre-training**
- **Transfer Learning**
- **Attention Mechanism**
- **Tokenization**
- **Fine-tuning**
- **Multilingual Capability**



Block diagram of Sentiment Analysis Using BERT



Example of BERT for Text Classification

TF-IDF

Term Frequency (tf): gives us the frequency of the word in each document in the corpus. It is the ratio of number of times the word appears in a document compared to the total number of words in that document. It increases as the number of occurrences of that word within the document increases. Each document has its own tf.

$$tf_{i,j} = \frac{n_{i,j}}{\sum_k n_{i,j}}$$

Inverse Data Frequency (idf): used to calculate the weight of rare words across all documents in the corpus. The words that occur rarely in the corpus have a high IDF score. It is given by the equation below.

$$idf(w) = log(\frac{N}{df_t})$$

Combining these two we come up with the TF-IDF score (w) for a word in a document in the corpus. It is the product of tf and idf:

$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

Let's take an example to get a clearer understanding.
Sentence 1 : The car is driven on the road.
Sentence 2: The truck is driven on the highway.
In this example, each sentence is a separate document.

We will now calculate the TF-IDF for the above two documents, which represent our corpus.

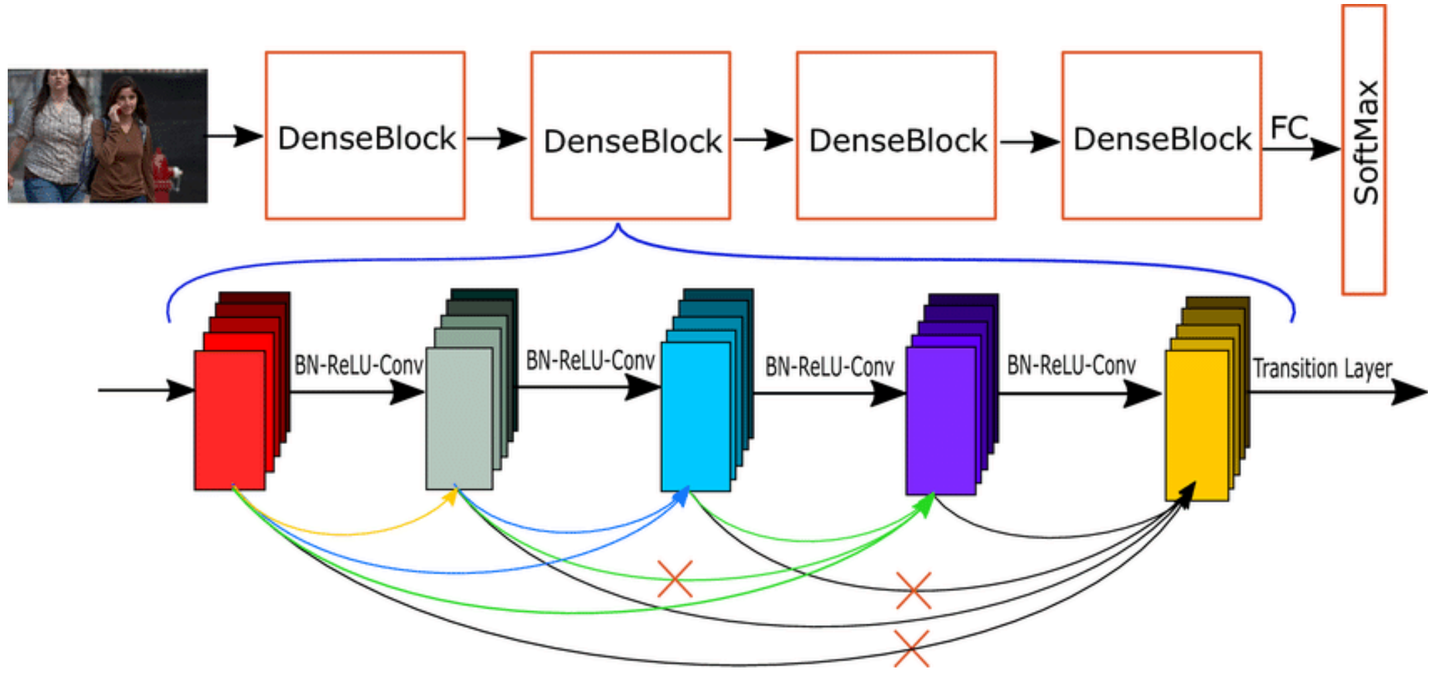
Word	TF		IDF	TF*IDF	
	A	B		A	B
The	1/7	1/7	log(2/2) = 0	0	0
Car	1/7	0	log(2/1) = 0.3	0.043	0
Truck	0	1/7	log(2/1) = 0.3	0	0.043
Is	1/7	1/7	log(2/2) = 0	0	0
Driven	1/7	1/7	log(2/2) = 0	0	0
On	1/7	1/7	log(2/2) = 0	0	0
The	1/7	1/7	log(2/2) = 0	0	0
Road	1/7	0	log(2/1) = 0.3	0.043	0
Highway	0	1/7	log(2/1) = 0.3	0	0.043

From this table, we can see that TF-IDF of common words was zero, which shows they are not significant. On the other hand, the TF-IDF of "car" , "truck", "road", and "highway" are non-zero. These words have more significance.

Generating Image Embeddings

DenseNet offers advantages such as efficient parameter utilization due to dense connections between layers, alleviating vanishing gradient issues, promoting feature reuse across layers, reducing the risk of overfitting, and achieving competitive or superior performance with fewer parameters compared to traditional architectures.

Advantages:-
1.Dense Connectivity
2. Optimal utilization of parameters
3. Addresses vanishing gradient problems
4. Reduced Overfitting
5. Compact Models
6. Feature Hierarchy



DenseNet which contains 4 dense blocks

EfficientNet

EfficientNet is a family of convolutional neural network architectures that optimally scales model depth, width, and resolution to achieve superior performance on image classification tasks. Through compound scaling, EfficientNet strikes a balance between accuracy and computational efficiency, making it a widely adopted choice in computer vision applications.

Advantages:-

1. Introduces compound scaling, efficiently balancing model depth, width, and resolution to optimize performance.
2. Achieving State-of-the-Art Performance by systematically scaling model parameters

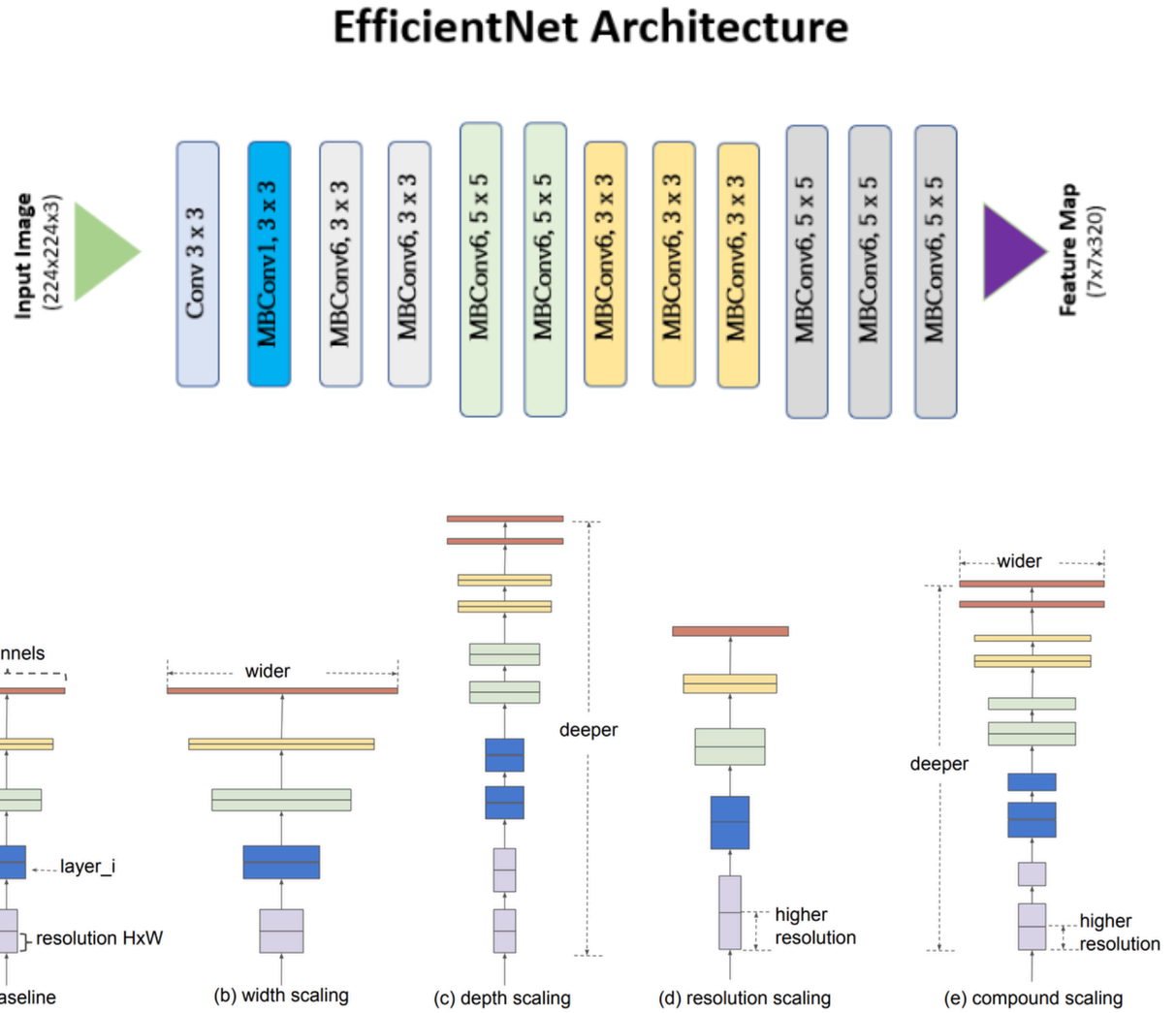


Figure 2. **Model Scaling.** (a) is a baseline network example; (b)-(d) are conventional scaling that only increases one dimension of network width, depth, or resolution. (e) is our proposed compound scaling method that uniformly scales all three dimensions with a fixed ratio.

RESULT

Initial Text Result

S. No	Model Used	Tf-IDF	BERT
1.	Logistic Regression	63%	59.4%%
2.	SVC	61.6%	59.6%
3.	MultinomialNB	62%	50.4%
4.	RandomForestClassifier	60%	56.6%
5.	XGBClassifier	63%	58%
6.	LGBMClassifier	56.7%	58.7%
7.	CatBoostClassifier	63.7%	59.6%

Initial Image Result

S.no	Model Used	EfficientNet	DenseNet
1.	Neural Network	45.6%	44.4%
2.	CatBoost	44.1%	43.3%
3.	XGBoost	42.7%	43.6%
4.	SVC	43.7%	41.6%
5.	RandomForest	43.7%	41.9%
6.	Logistic Regression	38.6%	37.9%

Initial MultiModal Results

Early Fusion

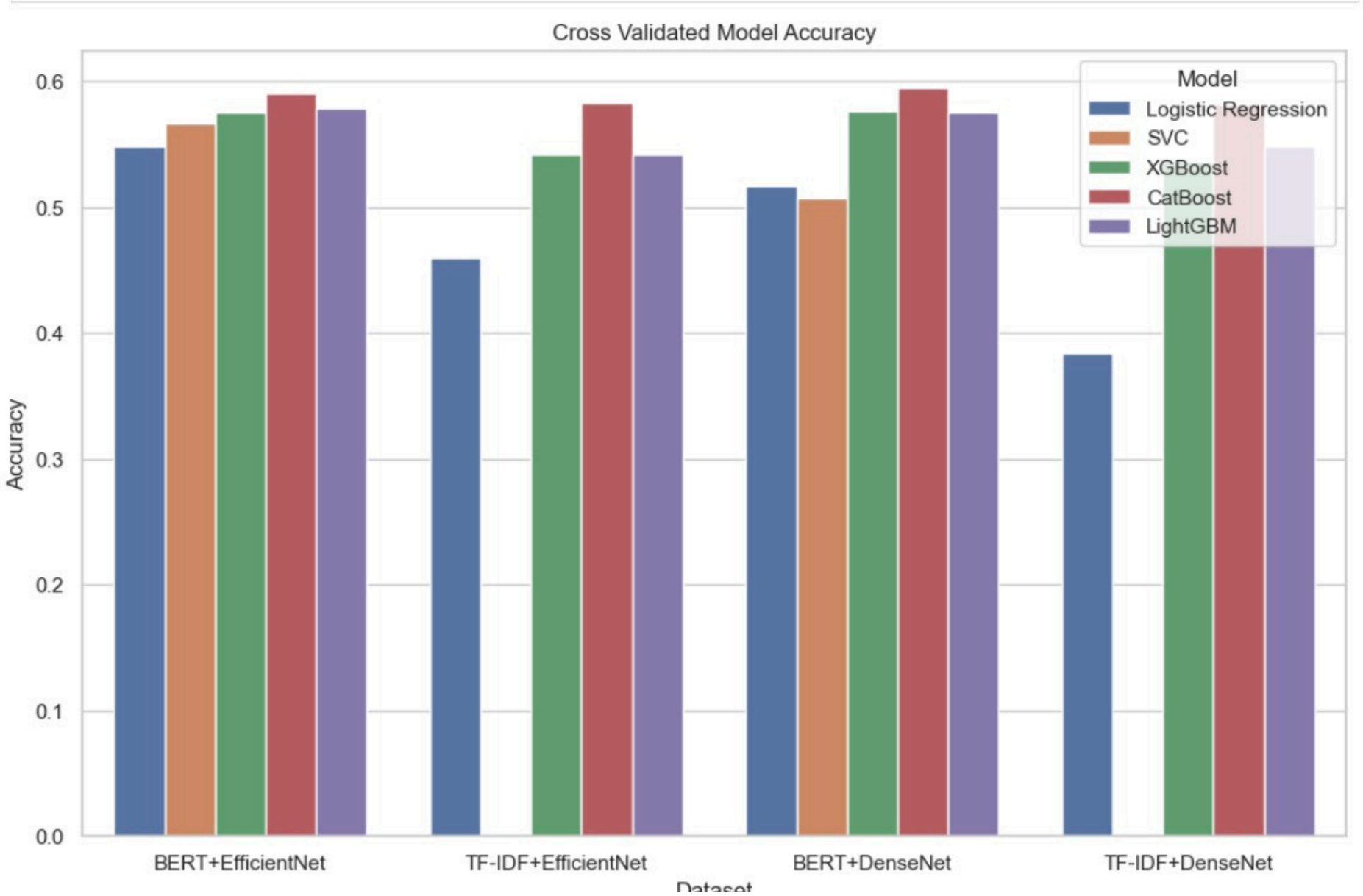
- **Definition:**
  Early Fusion in ML combines different data modalities at the model's early stage.

- **Implementation:**
  Text (BERT TF-IDF) and image (DenseNet, EfficientNet) embeddings concatenate early.

- **Model Processing:**
  Concatenation occurs before passing through subsequent layers.

- **Benefits:**
  Simultaneous consideration of information from both modalities.
  Enables joint learning and interactions early in the processing pipeline.

- **Contrast with Late Fusion:**
  Early fusion differs from late fusion, where modalities combine later in the model architecture.



Graph for Initial MultiModal Fusion Results

Our Proposed Model

Late Fusion

- **Late Fusion Overview:**
  Process information from different modalities separately until a specific point in the model.

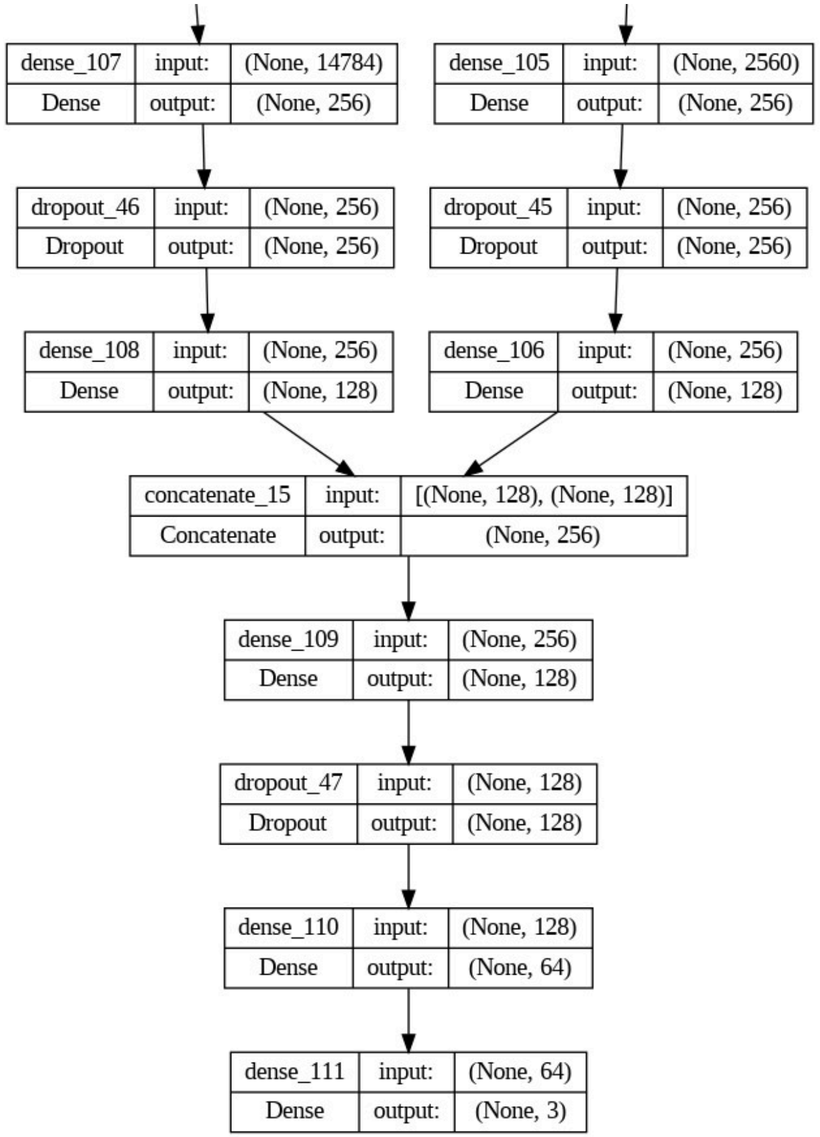
- **Embedding Processing:**
  Initial processing of image and text data through independent embedding layers.

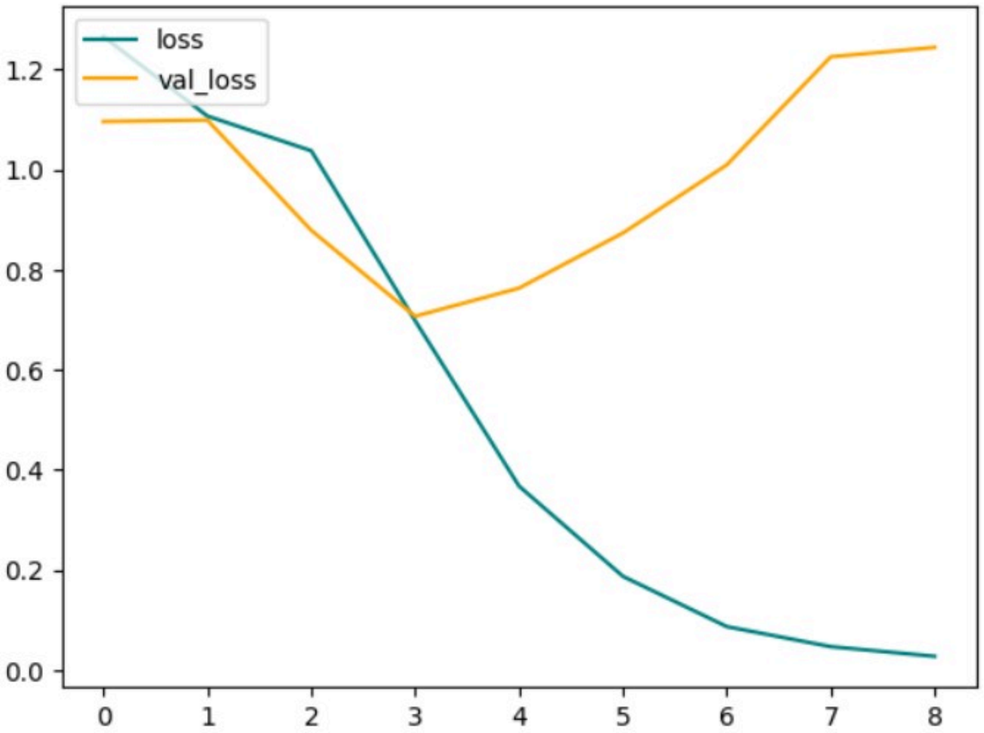
- **Concatenation Stage:**
  Concatenate representations of image and text after separate processing.

- **Complex Interactions:**
  Process combined representation through dense layers, capturing complex interactions between modalities.

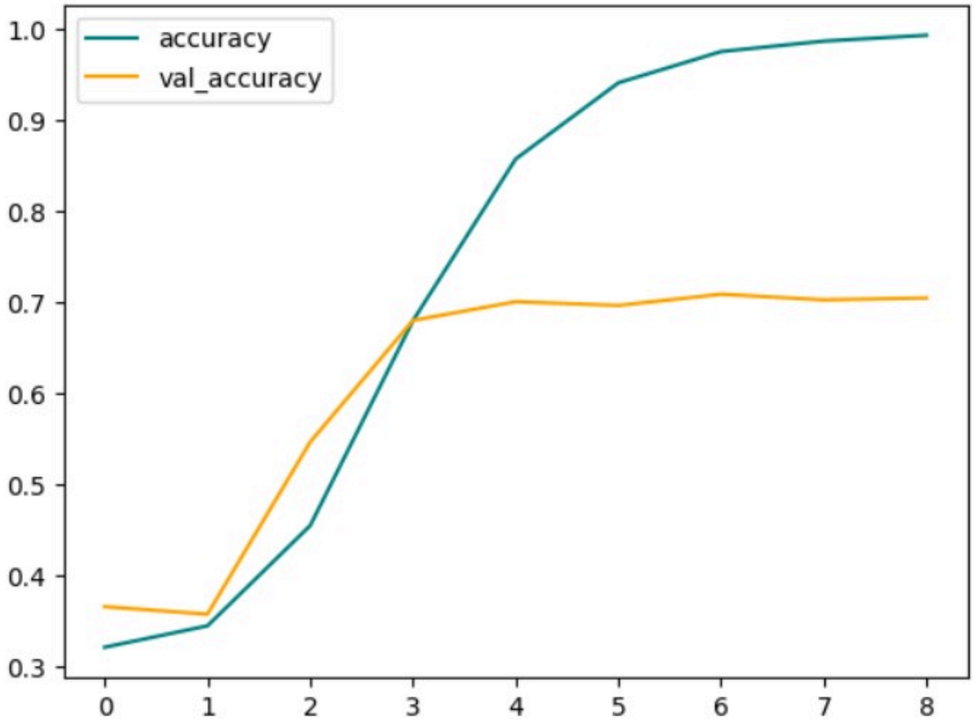
- **Performance Enhancement:**
  Leverage strengths of each modality for a more comprehensive representation, potentially improving overall model performance.

Model Architecture



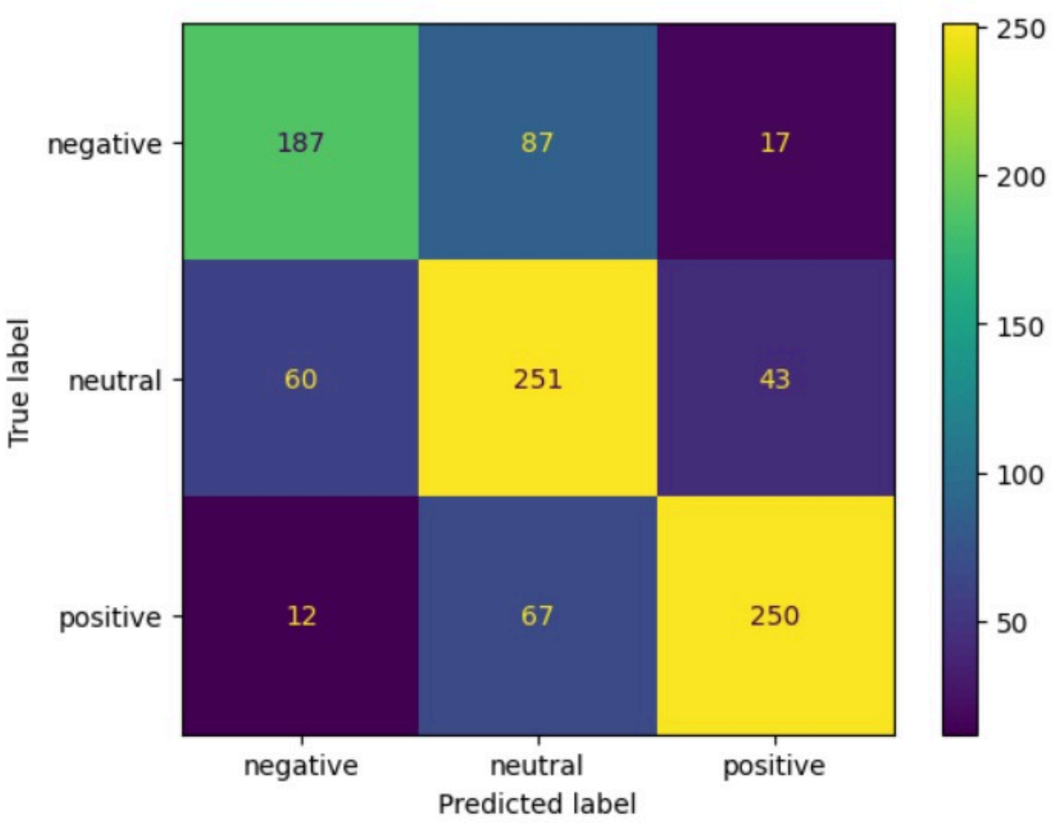


Loss

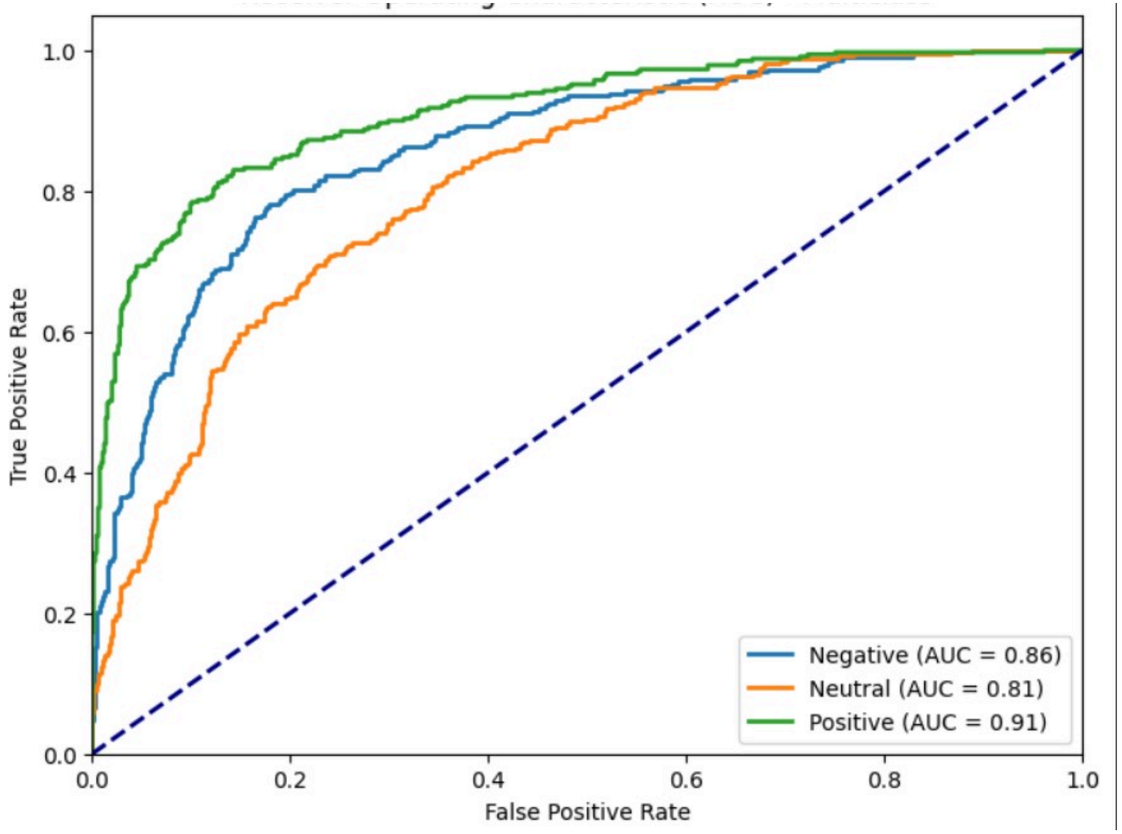


Accuracy

Cross Validated Accuracy-70.22%



Confusion matrix



Receiver Operating Characteristic(ROC)-Multiclass

Thank You